

MUHAMMAD AHMAD RATHORE, PH.D.

Assistant Professor · NUST-SEECS, Islamabad | Researcher — Cloud-Native Systems, Edge Computing, AI/ML Infrastructure

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RESEARCH PROFILE

Researcher and educator working across cloud-native edge computing, systems visibility, and AI-accelerated scientific computing. Ph.D. research at GIST (South Korea) focused on cloud-native edge orchestration and multi-view systems visibility; postdoctoral work at Northwestern University applied GPU-accelerated ML infrastructure to computational pathology. 8 Q1 journal publications, 8+ conference papers, 400+ citations, h-index 8, i10-index 5.

ACADEMIC APPOINTMENTS

Assistant Professor

Sept 2025 – Present

NUST – School of Electrical Engineering and Computer Science (SEECS), Islamabad

- Teach computer science courses with industrial-collaboration components spanning cloud infrastructure, Linux systems, and scalable service design.
- Conduct R&D aligned with department research priorities; affiliation verified on Google Scholar (seecs.edu.pk).

Adjunct Faculty

Fall 2024 – Spring 2025

University of New Hampshire, Department of Computer Science

- Taught CMPL 530 (Python Programming I) and CMPL 637 (Python Programming II) to classes of approximately 30 students.
- Developed lab curriculum covering automation, infrastructure scripting, and data handling with NumPy/pandas.

Scientific Officer / Lecturer

Jan 2016 – Feb 2017

COMSATS Institute of Information Technology

- Taught Software Engineering, Network Security, and Object-Oriented Programming at the undergraduate level.
- Supervised undergraduate final-year projects and the IEEE Computer Society student chapter, COMSATS Wah campus.

POSTDOCTORAL RESEARCH

Postdoctoral Fellow

Mar 2022 – Jan 2024

Northwestern University, Feinberg School of Medicine

- Built a GPU-accelerated, cloud-based framework with Kitware, Inc. for whole-slide pathology image analysis, under an NIH National Cancer Institute-funded collaboration (grant 5U01CA220401-04) with Wake Forest School of Medicine and Emory University.
- Co-authored HistomicsStream, a whole-slide-image TensorFlow reader delivering a 65% runtime performance improvement; contributed to HistomicsML, HistomicsUI, and HistomicsTK.
- Optimized AI inference and MLOps pipelines using NVIDIA DALI, cuCIM, Triton Inference Server, and TensorRT.

Postdoctoral Fellow

Sep 2021 – Feb 2022

GIST Blockchain Economy Center, South Korea

- Developed open-source interfaces for Uniswap-style decentralized token exchange (GISTSWAP, DelioswapV3 custom contracts).
- Technical stack: AWS (S3, EC2), TypeScript, JavaScript, Python, Solidity, Rust.

DOCTORAL RESEARCH

Ph.D., Computer Science — Networked Intelligence Lab

Mar 2017 – Aug 2021

Gwangju Institute of Science and Technology (GIST), South Korea — Fully funded

- Doctoral thesis on forecasting SmartX multi-view spatio-temporal visibility for cloud-native edge computing boxes.

- Technical member on Asi@Connect / TEIN federated cloud-native edge infrastructure projects spanning 10 countries; Kubernetes and OpenStack operations.
- Conducted the OF@TEIN++ Annual Project Conference 2021, inviting 50+ guests; supported project funding of €50,000 as project support staff, assisting the PI with proposals and financial/interim/final reporting.

EDUCATION

Ph.D., Computer Science

2017 – 2021

Gwangju Institute of Science and Technology (GIST), South Korea

M.Sc., Information & Communication Systems Security

2008 – 2010

KTH Royal Institute of Technology, Sweden — Network security and cryptography

B.S. (Hons), Computer Science / Software Engineering

COMSATS Institute of Information Technology

SELECTED PUBLICATIONS

8 Q1 journals · 8+ conference papers · 400+ citations · h-index 8 · i10-index 5 (Google Scholar, verified)

Amgad, M., Rathore, M. A., Cooper, L. "A panoptic segmentation approach for tumor-infiltrating lymphocyte assessment: development of the MuTILs model and PanopTILs dataset." *npj Breast Cancer (Nature)*, 2024.

Risdianto, A. C., Usman, M., Rathore, M. A. "Transforming Network Management: Intent-Based Flexible Control Empowered by Efficient Flow-Centric Visibility." *Future Internet* 16.7 (2024): 223.

Zeb, S., Rathore, M. A., et al. "Towards AI-enabled NextG Networks with Edge Intelligence-Assisted Microservice Orchestration." *IEEE Network*, Feb 2023.

Farasat, T., Rathore, M. A., Khan, A., et al. "Machine Learning-based BGP Traffic Prediction." Conference paper, Nov 2023.

Farasat, T., Rathore, M. A., Asim, Z., et al. "Poster: SmartX BGP BVT: A First Real-Time BGP Blackholing Visibility Tool." *ACM IMC* 2023.

Ahmad, M., Kim, J. "Spatio-temporal summarized visualization of SmartX multi-view visibility in cloud-native edge boxes." *Computer Science and Information Systems*.

Rathore, M. A., Usman, M., Kim, J. "Short paper." *Transactions on Emerging Telecommunications Technologies*, Vol. 32, Issue 6, June 2021.

Rathore, M. A., Usman, M., Kim, J. "Conference paper." *CFI 2018: Proceedings of the 13th International Conference on Future Internet Technologies*.

Mamun, M. S. I., Rathore, M. A., Lashkari, A. H., Stakhanova, N., Ghorbani, A. A. "Detecting malicious URLs using lexical analysis." *International Conference on Network and System Security*, 467–482, 2016. (344 citations)

Full publication list: scholar.google.com/citations?user=GoiH9mkAAAAJ

RESEARCH INTERESTS

Cloud-native systems · Edge computing · Machine vision · High-performance computing · Systems monitoring and management · Cybersecurity

PROFESSIONAL CERTIFICATIONS

Certified Ethical Hacker (CEH), EC-Council · CCNA · Workshop Series for New Managers